

Quadratic Residues

Pushpendra Pal

1 Legendre Symbol

- p is odd prime
- a s.t. $\text{GCD}(a,p) = 1$
- $\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is QR mod } p \\ 0 & \text{if } a \text{ is QNR mod } p \end{cases}$
 $a \neq 0$
- $\left(\frac{a}{p}\right) = a^{(p-1)/2} \pmod{p}$

Examples. $\left(\frac{4}{11}\right) = 4^{(11-1)/2} = 1$, hence there is a solution to $x^2 \equiv 4 \pmod{11}$. Use Tonelli Shanks algorithm to find x .

$\left(\frac{5}{7}\right) = -1$, hence there is no solution.

- $\left(\frac{a}{p}\right) \cdot \left(\frac{b}{p}\right) = \left(\frac{a \cdot b}{p}\right)$

2 Jacobi Symbol

- Let $b \equiv 1 \pmod{2}$, $a \in \mathbb{Z}$ and $b = p_1^{k_1} \cdot p_2^{k_2} \dots p_n^{k_n}$, then
- $\left(\frac{a}{b}\right) = \left(\frac{a}{p_1}\right)^{k_1} \left(\frac{a}{p_2}\right)^{k_2} \dots \left(\frac{a}{p_n}\right)^{k_n}$

3 Some thms and lemmas on legendre

- **Wilson's Theorem**

$-(p-1)! \equiv -1 \pmod{p}$. Prove using units.

- **Count of QRs and QNRs**

$\exists \frac{p-1}{2}$ QRs and $\frac{p-1}{2}$ QNRs.

- **Is $p-1$ a QR?**

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases},$$

so let's say we have $p=13$, then $x=12$ (-1) is a QR, no need to check. But if $p = 19$, then $x=18$ (-1) is not a QR automatically.

- $\left(\frac{2}{p}\right) = 1^{\frac{p^2-1}{8}}$

- **Quadratic reciprocity**

- If p, q are distinct odd primes, what is the relation between q being a QR mod p and p being a QR mod q .

- It says that the congruences are either both solvable or both unsolvable, unless $p \equiv q \equiv 3 \pmod{4}$, in which case, one is solvable while the other is not.

- IF $p \equiv q \equiv 3 \pmod{4}$, $\left\{ \left(\frac{q}{p}\right) = -\left(\frac{p}{q}\right) \right\}$

ELSE $\left\{ \binom{q}{p} = \binom{p}{q} \right\}$.

Either both are equal -1 or both are equal to 1, when p and q are both not 3 mod 4.

- Combining both,

$$\binom{q}{p} \cdot \binom{p}{q} = -1^{\frac{p-1}{2} \frac{q-1}{2}}$$

- **Gauss's lemma** to calculate $\binom{a}{p}$, $\text{GCD}(a,p) \equiv 1$

- Take the least residues of $S = \{a, 2a, \dots, \left(\frac{p-1}{2}\right)a\}$

- $S \subseteq \{0, \dots, p-1\}$

- Let $u = \#$ members of $S > p/2$.

- Then, $\binom{a}{p} = (-1)^u$

- There are only 0 or 2 solutions if $p \nmid a$. p being an odd prime. If p is composite, then there may be multiple solutions to $x^2 = a \pmod{p}$. See an example in next section.

4 QRs on multiple prime composites

- If $n = p_1^{r_1} \cdot p_2^{r_2} \dots p_k^{r_k}$, then relative primality implies that it's enough to solve the congruences $x^2 = a \pmod{p_i^{r_i}}$ for each i

- Example: Solve $x^2 \equiv 79 \pmod{91}$.

This is equivalent to $x^2 \equiv 79 \pmod{13} \equiv 1 \pmod{13}$ and $y^2 \equiv 79 \pmod{7} \equiv 2 \pmod{7}$

$$\begin{aligned} x^2 &\equiv 1 \pmod{13} & y^2 &\equiv 2 \pmod{7} \\ x &\equiv 1, 12 \pmod{13} & y &\equiv 3, 4 \pmod{7} \end{aligned}$$

, Now we use Chinese Remainder Theorem (CRT) over different pairs with one from each side $\{1, 12$ or $-1\}$ and $\{3, 4\}$.

Remember CRT on $(a \pmod{p})$ and $(b \pmod{q})$ works like this,

$$v = (a \cdot q \cdot (q^{-1} \pmod{p}) + b \cdot p \cdot (p^{-1} \pmod{q})) \pmod{pq}$$

$$\begin{aligned} (1, 3) &\rightarrow 1 \cdot 7 \cdot (7^{-1} \pmod{13}) + 3 \cdot 13 \cdot (13^{-1} \pmod{7}) &\rightarrow 1 \cdot 7 \cdot 2 + 3 \cdot 13 \cdot 6 &\rightarrow 14 + 234 &\rightarrow 240 \equiv 66 \pmod{91} \\ (1, -3) &\rightarrow 1 \cdot 7 \cdot (7^{-1} \pmod{7}) + (-3) \cdot 13 \cdot (13^{-1} \pmod{7}) &\rightarrow 1 \cdot 7 \cdot 2 + (-3) \cdot 13 \cdot 6 &\rightarrow 14 - 234 &\rightarrow -220 \equiv 53 \pmod{91} \\ (-1, 3) &\rightarrow (-1) \cdot 7 \cdot (7^{-1} \pmod{7}) + 3 \cdot 13 \cdot (13^{-1} \pmod{7}) &\rightarrow (-1) \cdot 7 \cdot 2 + 3 \cdot 13 \cdot 6 &\rightarrow -14 + 234 &\rightarrow 220 \equiv 38 \pmod{91} \\ (-1, -3) &\rightarrow (-1) \cdot 7 \cdot (7^{-1} \pmod{7}) + (-3) \cdot 13 \cdot (13^{-1} \pmod{7}) &\rightarrow (-1) \cdot 7 \cdot 2 + (-3) \cdot 13 \cdot 6 &\rightarrow -14 - 234 &\rightarrow -240 \equiv 25 \pmod{91} \end{aligned}$$